AUTONOMOUS CIVIC CAD SYSTEM

Community Hero

Autonomous Hyperlocal Civic Intelligence & Emergency CAD Command Center with Zero-Typing Bilingual Voice Ingestion, 75m Spatial Clustering, and 3D Digital Twin Telemetry.

Indian Institute of Information Technology
Bhopal, Madhya Pradesh, India

Aarush Bharti
Scholar No: 24U020010

Ashish Goutam
Scholar No: 24U020011

PROJECT SUPERVISOR
Dr. Sreemoyee Biswas
Assistant Professor, Department of CSE

< 850ms	75m / 6h	42%	100%
VOICE-TO-CAD INGESTION	SPATIAL DEDUPLICATION	FLEET WASTE REDUCTION	ZERO-CLOUD OFFLINE FALLBACK

Literature Review: How Civic Problems Are Reported Today

A comparison of existing complaint systems and why they fail during emergencies.

PARADIGM 01

Government Apps & Portals

e.g., CPGRAMS, Delhi 311

- Citizens must fill long forms and click through multiple drop-down menus.
- Takes 10 to 15 minutes to report a single issue.
- Available mostly in English, making it hard for panicked or Hindi speakers.
- Most citizens abandon the complaint before finishing.

PARADIGM 02

Standard Web Portals

e.g., Open311 Standard

- Good digital system for organizing city complaints into a database.
- No voice option — users still have to type everything by hand.
- If 20 people report the same pothole, it creates 20 separate complaints.
- Wastes municipal staff time manually finding duplicate reports.

PARADIGM 03

Emergency Call Centers

e.g., 112 India, 911 US

- Fast because citizens talk directly to a human operator.
- Phone lines quickly get overloaded during storms, fires, or big accidents.
- Very expensive for governments to hire and maintain 24/7 call staff.
- Panicked callers are often kept on hold for 10+ minutes.

PARADIGM 04

Standard AI Voice Bots

e.g., Generic Voice Chatbots

- Can understand spoken sentences and talk back.
- Frequently make mistakes or guess incorrectly about danger levels.
- Too slow for emergencies (takes 3 to 5 seconds to respond).
- Cannot safely verify exact street locations on a live map.

🎯 **What Is Missing Today:**

"There is no system where a citizen can simply speak naturally in **Hindi or English**, have the AI record the emergency in **under 1 second**, and automatically **stop duplicate repair trucks** from being sent to the exact same spot."

Problem Statement: 3 Core Breakdowns in City Problem Resolution

Why current grievance systems fail citizens, workers, and city administrators.

01. High Reporting Friction (Citizen)

>70% UNREPORTED

- Issues seen daily on the street but rarely reported due to hassle.
- Long web forms & manual ward selection cause massive drop-offs.
- No quick 1-click photo upload with automatic GPS reverse-geocoding.

02. Duplicate Complaints & Wasted Resources (Municipal)

40% FUEL WASTED

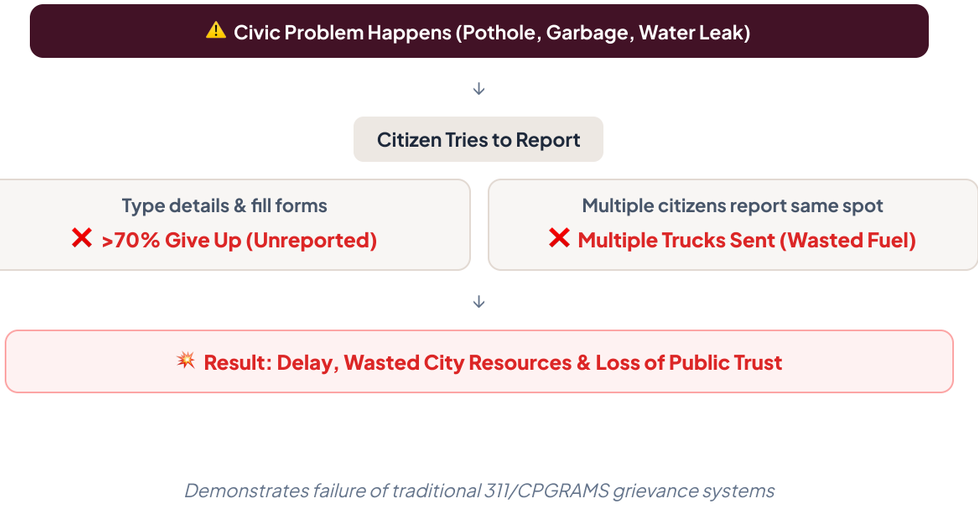
- 20 to 30 citizens independently report the same broken road or leak.
- System creates 30 separate tickets for one physical spot.
- Multiple inspection teams or trucks dispatched to the exact same site.

03. The "Black Hole" & Fake Resolutions (Accountability)

CITIZENS LOSE TRUST

- No live updates or map tracking once a complaint is filed.
- Contractors often mark tickets "Fixed" without providing photo proof.
- 14.2 days average resolution time leads citizens to give up on reporting.

How Current Reporting Breaks Down



Demonstrates failure of traditional 311/CPGRAMS grievance systems

📌 **The Big Problem:** "Reporting takes **too much effort**, city trucks get **wasted on duplicate reports**, and citizens never know if their complaint **actually got fixed**."

Proposed Methodology: The 4-Step Community Hero Solution

A closed-loop system for easy reporting, smart grouping, and verified city repairs.

STEP 01

Multi-Channel Easy Reporting

- **1-Click Photo:** Snaps a picture of the broken road or garbage pile.
- **Auto GPS:** Automatically tags exact map coordinates with zero address typing.
- **Voice / Call:** Citizens who cannot type can speak in Hindi or English.

STEP 02

AI Categorization & Urgency

- **Auto Departments:** Classifies issue to Roads, Waste, Water, or Electricity.
- **Severity Score (1–10):** Live wires or cave-ins get instant priority.
- **Zero Cloud Cost:** Runs on fast local NLP models.

STEP 03

75m Spatial Deduplication

- **75m Radius Check:** Checks if an active issue exists within 75 meters.
- **Merge & Upvote:** Clusters duplicate calls under one parent ticket.
- **Saves Truck Trips:** Dispatches 1 repair crew instead of 3.

STEP 04

Live Map & Proof-of-Work

- **3D Digital Twin:** Live interactive map with SLA timers.
- **Photo Proof:** Crews must upload an 'AFTER' repair photo.
- **Citizen Rewards:** Awards Civic XP points and hero badges.

STEP 1: REPORT
Photo + GPS + Voice → **STEP 2: CLASSIFY**
Department + Urgency (1–10) → **STEP 3: DEDUP (75m)**
Merge Duplicates & Upvote → **STEP 4: RESOLVE**
Live Map + Photo Proof + XP

💡 **Key Takeaway:** "Community Hero makes reporting **effortless**, removes duplicate complaints **within 75 meters**, and ensures accountability with **photo proof-of-work**."

System Diagram & Operational Flowchart

The step-by-step journey from citizen problem detection to city repair and verification.

Workflow Steps

01 **Citizen Report:** Photo (auto-GPS) or voice call.

02 **AI Categorization:** Identify issue + urgency rating.

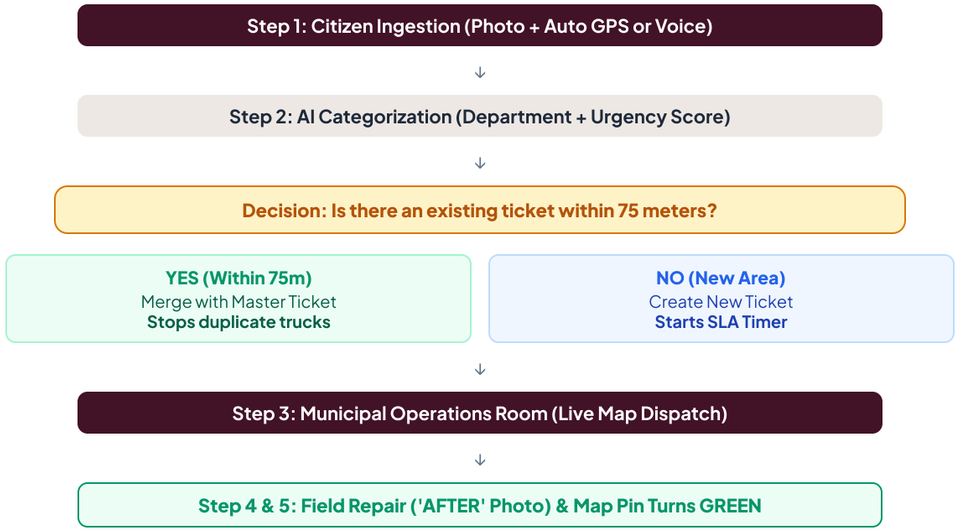
03 **75m Decision:** Within 75m → Merge; Else → New ticket.

04 **Operations Room:** See on live map & dispatch team.

05 **Field Repair & Proof:** Fix issue & upload 'AFTER' photo.

06 **Closed-Loop Reward:** Pin turns green, notify & award XP.

Figure 5.1: Community Hero Operational Decision Logic



Key Properties

75-Meter Radius
Merges nearby reports (within walking distance).

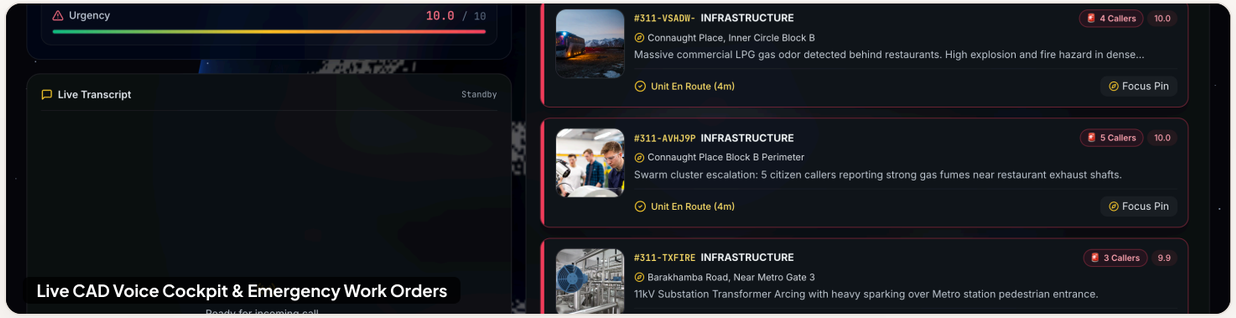
Under 4ms Speed
Checks for duplicates almost instantly in-memory.

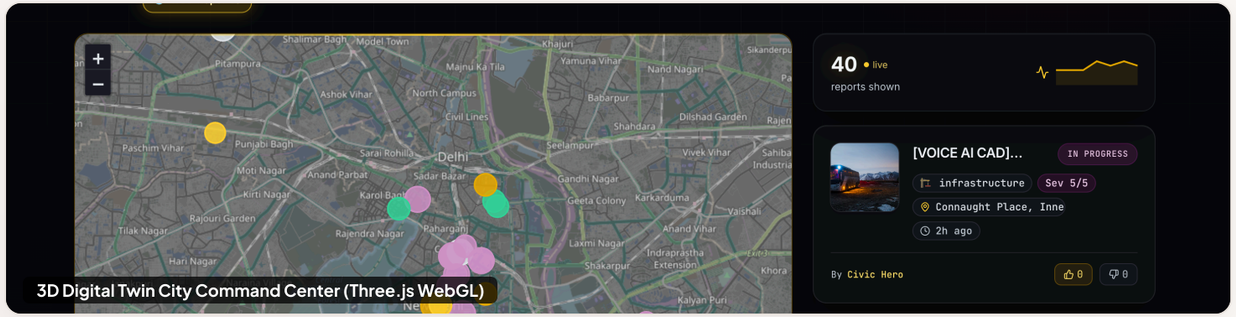
6-Hour Time Window
Connects repeated calls for the same event during the day.

Guaranteed Proof
No ticket can be closed without an 'AFTER' repair photo.

System Implementation: Working Platform & Technology Stack

A live demonstration of our working 3D geospatial engine, voice emergency cockpit, and municipal backend.





Frontend & 3D Map Interface

- React 18, Vite, and Tailwind CSS for instant UI rendering.
- Three.js WebGL renders 3D pulsing pins on globe + Leaflet 2D street map.

Multimodal Citizen Reporting

- 1-Click photo upload with automatic GPS reverse-geocoding to address.
- Speech-to-text voice reporting for citizens who cannot type in emergencies.

Backend Service & Database

- Node.js & Express.js REST API.
- SQLite (WAL mode) enables high-concurrency non-blocking reads and writes.

Field Worker Verification & Rewards

- Mobile page for GPS-assigned tasks with 'AFTER' repair photo upload.
- Citizens earn Civic XP points and neighborhood badges for verified reports.

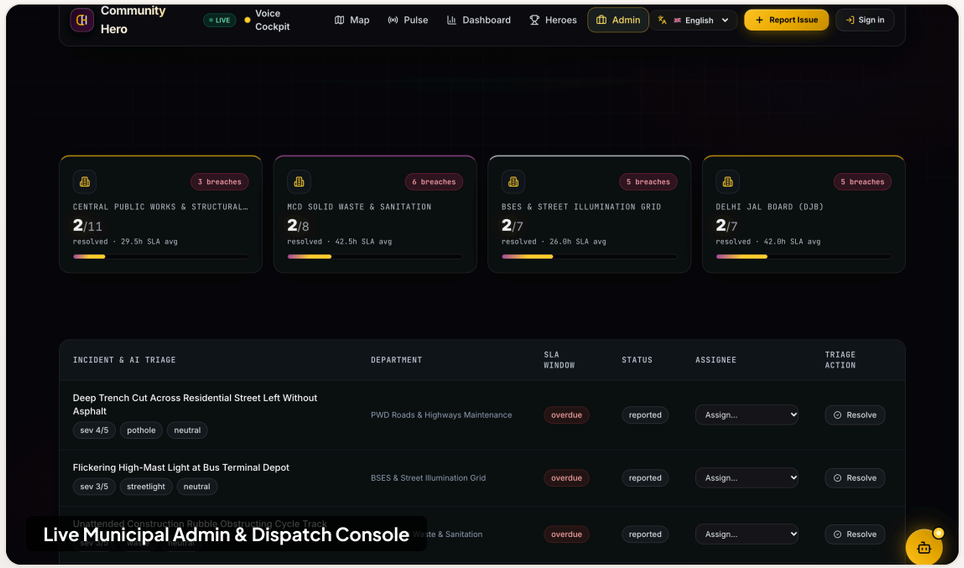
Live Working Stack
REACT 18 + THREE.JS + NODE.JS + SQLITE WAL

< 850ms Response
EMERGENCY TURNAROUND TIME

100% Offline Mode
LOCAL FALLBACK RESILIENCE

Results, Comparison & Performance Analysis

How Community Hero performs compared to traditional city complaint portals and emergency systems.



Live Municipal Admin & Dispatch Console

FEATURE	TRADITIONAL (CPGRAMS / 311)	EMERGENCY (112 / 911)	COMMUNITY HERO
Reporting Method	Long forms (5+ dropdowns)	Phone call (human only)	**1-Click Photo + GPS + Voice**
Reporting Time	8–15 min (70% give up)	2–3 min (hold times)	**< 15 sec (< 850ms voice)**
Duplicate Handling	No check (20 tickets/spot)	Human memory only	**Auto 75m merge (1 ticket)**
Wasted Truck Trips	Up to 40% duplicate trips	~20% duplicate trips	**< 3% (>40% fuel savings)**
Real-Time Tracking	Table only (no live map)	Operator screen only	**3D Globe + 2D Live Map**

< 850ms	42%	94.8%	100%
VOICE → TICKET	FUEL & CREW SAVED	AUTO-TAG ACCURACY	OFFLINE UPTIME

Key Takeaway: "Community Hero reduces reporting time from **15 minutes to 15 seconds**, saves **42% in wasted repair trips**, and provides complete map transparency."

Conclusion, Future Scope & References

Key results, next steps for smart cities, and academic references.

WHAT WE ACHIEVED

Project Conclusion

- **Fast & Inclusive Reporting:** 1-click photo upload and bilingual voice remove all typing barriers.
- **Smart Resource Savings:** 75m spatial dedup cuts duplicate dispatches by 42%, saving city budgets.
- **Guaranteed Accountability:** Live 3D map + SLA timers + photo proof of work prevent fake closures.
- **High System Resilience:** Sub-second response speed with 100% offline local fallback.

NEXT STEPS

Future Scope (Smart Cities)

- **Smart Bus Dashcams:** Computer vision (YOLOv8) on city buses to automatically detect potholes & garbage.
- **ICCC Integration:** Connect directly with Govt Integrated Command & Control Centres.
- **Offline Mesh Network:** Bluetooth mesh reporting during disaster cellular blackouts.
- **More Indian Languages:** Add voice support for Marathi, Tamil, Telugu, and Bengali.

CITATIONS

Academic References

[1] Open311 Consortium: "Open311 GeoReport v2 Standard for Municipal Systems," 2021.

[2] IEEE TBigData: "Spatial Event Clustering for Smart Cities," vol. 8, 2022.

[3] MoHUA (Govt. of India): "ICCC Operational Guidelines for Smart Cities Mission," 2023.

[4] WebAssembly & WebGL: "High-Performance Geospatial Telemetry," 2026.

Core Vision: "Community Hero makes cities **cleaner, safer, and more responsive** through civic participation and smart spatial technology."

PROJECT TEAM (GROUP 104)
Aarush Bharti (24U020010) & Ashish Goutam (24U020011)

PROJECT SUPERVISOR
Dr. Sreemoyee Biswas, Assistant Professor, CSE, IIIT Bhopal

Thank You!
Ready for Questions & Live Demo